Mechanical Lab Safety Report

Build and Testing of a 3D Hologram-like LED Fan Display

Date:	21/07/2026
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Student contact nr:	0794486205
Supervisor:	Dr Melody Neaves
Lab engineer:	Mr Kevin Neaves
Head of safety:	Mr Cobus Zietsman

Emergency Contacts:

Contact:	Room nr.	Work nr.	Cell nr.
Mr Kevin Neaves	M3002	-	-
Mr Cobus Zietsman	M212	021 808 4275	-
Campus Security	-	021 808 2333	WhatsApp: 082 808 2333
Fire Brigade	-	021 808 8888	-
Ambulance	-	021 883 3444	-

Signatures:

Student:
(Teagan Reddy)
Supervisor:
(Dr Melody Neaves)
Lab Engineer:
(Mr Kevin Neaves)
Head of Safety:
(Mr Cobus Zietsman)

27-7-2026

Pressure vessels or pipes (check relevant box):

☒ No pressure vessels or pipes with pressure in excess of 50kPa are involved in this project.

☐ Pressure vessels or pipes in excess of 50kPa are involved – additional signature and report required (refer to Safety Report Guidelines on STEMLearn).

Hot work / working at heights / confined entry / excavation (check relevant box):

☒ No hot work / working heights / confined entry / excavation work involved in this project.

☐ Hot work / working heights / confined entry / excavation work (underline relevant work type(s)) involved in this project – additional signature and report required (refer to Safety Report Guidelines on STEMLearn).

Document Version History

This table records the revision history of this safety report, which will be updated whenever the report is revised. For example, when the final design is built, when equipment details are confirmed, or when new activities or risks are introduced. Furthermore, the linear rail system design has not yet been finalised, the associated risks will be incorporated in a future revision of this report.

Version	Date	Author	Description of Changes
1.0	15/06/2026	Teagan Reddy	Initial concept-stage safety report. Based on CAD concept models only, no physical final design yet. Covers planned testing activities, general lab safety, fire safety and the activity based risk assessment.
2.0	18/06/2026	Teagan Reddy	Minor improvements to Detailed Experimental Procedure and General Housekeeping sections.
3.0	21/07/2026	Teagan Reddy	Added full detail design section with specific components.

Overview of Testing

Type of test and standard (if applicable):

Test type	Functional testing of a 3D hologram-like LED fan display
Standard(s)	No formal material-test standard applies. Design references: IEC 60529 (finger-safety probe, guard openings) and SANS 10142 / general electrical-safety good practice.

Equipment to be used:

Equipment type	Make & model	Measurement range (if applicable)	Resolution (if applicable)
DC Power Supply	GW Instek SPD-3606	0 to 30 V 0 to 5A	-
Multimeter	DT-9205A	0 to 200 V 0 to 5A	-
Holographic Fan Unit	Custom	-	-
LED Strip	APA102	5V, 3A at full brightness	-
LiPo Battery	Generic	5000mAh 3.7V	-

Detailed Experimental Procedure

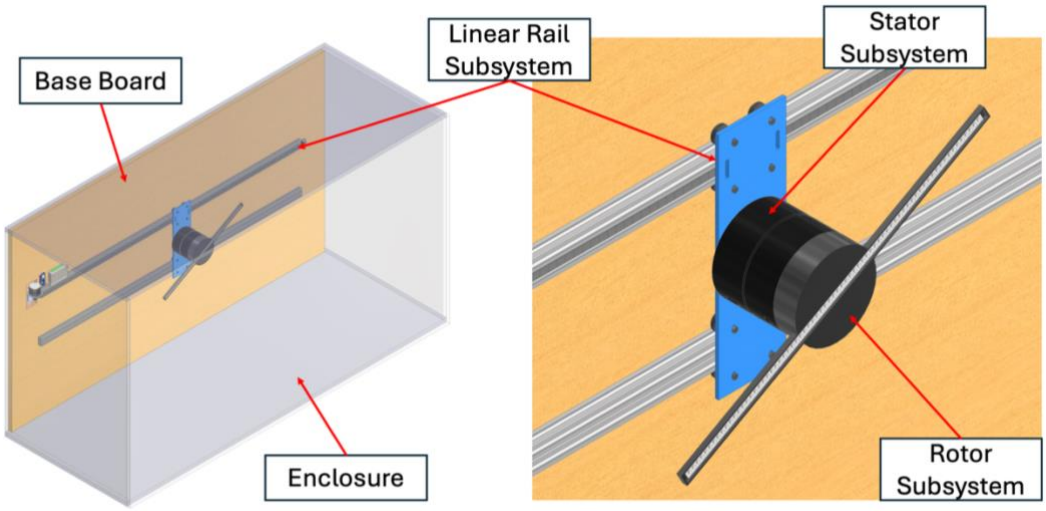


Figure 1: Full system CAD model

The fan will be powered initially from the bench DC power supply (rather than its final integrated supply) so that the maximum voltage and current to the motor can be limited for safety during early testing. As seen in Figure 1, the system has a rotating blade and a linear rail system, which presents multiple pinch and contact risks. Therefore, the entire assembly will be enclosed in a cardboard/plastic box. Mr Kevin Neaves or Mr Cobus Zietsman must inspect and approve the assembly before it is run for the first time, to confirm it is stable and safe. Nearby laboratory users will be warned before every start, told of the risks (flying debris if a part detaches) and asked to move to a safe distance. If at any point the fan becomes unstable or unsafe, it will be shut down immediately. Testing only resumes once the design has been corrected.

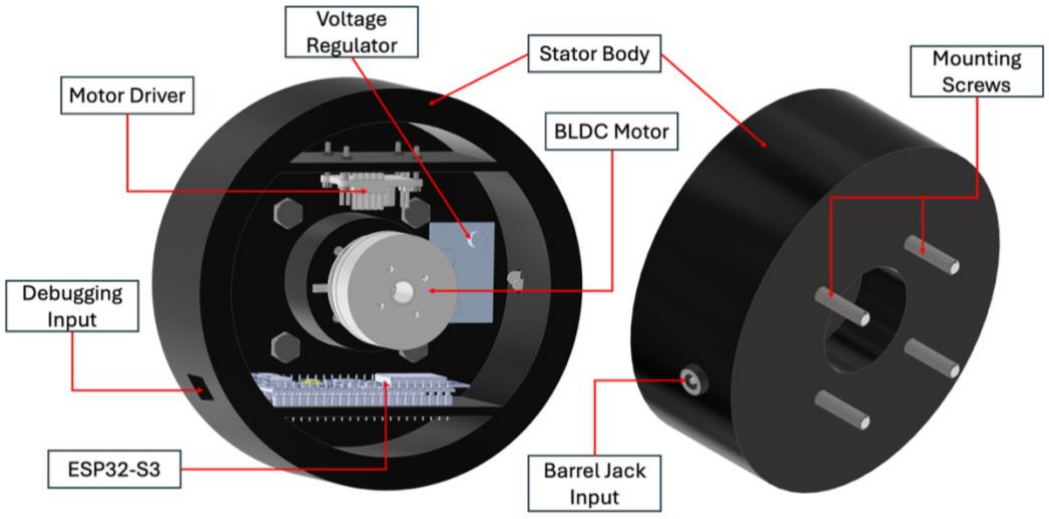


Figure 2: Stator Subsystem CAD model

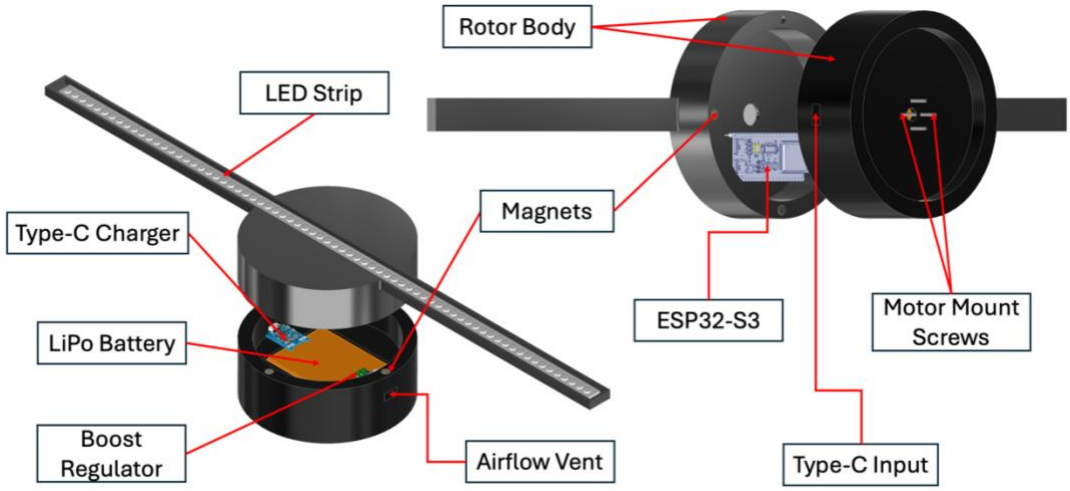


Figure 3: Rotor Subsystem CAD model

Figures 2 and 3 show the detailed overview of the rotor and stator subsystems. The addition of a LiPo battery introduces an electrical and fire risk. The system is mounted and held together by screws and magnets.

Order of operating steps

1. Assemble the fan without the LED arm blades attached and connect it to the bench DC power supply set to 12 V, current-limited (e.g. 3 A).
2. Place the protective box around the system, with a small opening at the front to observe the display.
3. Safety glasses must be obtained from the workshop and used before powering on and enabling the DC supply output.
4. Before the first start, Mr Kevin Neaves or Mr Cobus Zietsman must be present to confirm the fan is safe and stable enough to run.
5. Start the rotor slowly at low speed (50 rpm) using the microcontroller, with no arm attached. The rotor speed is verified through feedback from the closed-loop motor control and the digital hall effect sensor, to ensure the system is operating as intended.
6. Once stable operation is confirmed (no major vibration or safety concerns), slowly increase speed toward the 700 rpm target, as defined in the system specifications.
7. Stop the fan and attach/secure the LED arm assembly.
8. Repeat the low-speed-to-target-speed checks with the arm attached, to confirm it adds no instability. The system will be visually inspected to monitor any vibrations or instability at low speed.
9. Add the LEDs and remaining electronics to the rotating assembly and repeat the checks.
10. Record/monitor the displayed images and scenes through the viewing opening.
11. Once testing on the bench DC supply is complete, disconnect it. Before connecting the LiPo battery for final integrated-supply operation, verify its voltage with a multimeter to confirm it is within the safe operating range.
12. Once all testing is complete and the fan has stopped fully, disconnect all power sources and disassemble the setup safely.

Warning Symbols



(a) Flying Debris



(b) Epilepsy Risk



(c) Spinning Blades



(d) Soldering Station



(e) Electrical Hazard



(f) Pinch Point

Figure 1: Applicable warning/hazard symbols

General Laboratory Safety

The following general laboratory safety instructions are applicable:

- No afterhours testing may be performed without the necessary permissions[1].
- Full supervised training is required before testing may be undertaken. Permission to proceed with unsupervised testing should be signed off by the lab engineer.
- Closed shoes must be worn at all times.
- Emergency equipment must be located and easily accessible.
- Students may not work alone in the laboratory.
- Emergency exits must be known. The nearest exits applicable to the setup are provided in Appendix A of this document.
- Loose clothing may not be worn. Loose hair must be tied up.

[1] Written permission from supervisor and approved by the chief safety officer. Attach proof of this permission to this document and note here the times and conditions of the arrangement.

- Good housekeeping practices should be maintained during testing. The lab should be completely clean, including all equipment stored away, after testing. Refer to the General Housekeeping section for particulars regarding practices to be followed for this specific setup.
- No food or drink is permitted in the laboratory.
- Safety report must be visible and accessible during testing.
- No equipment or test may be left unattended.

Anticipated Interactions with other Laboratory Users

Testing will be conducted at a single workstation in M214, the M&M Training Workshop. Other users of the workshop will be warned whenever the fan is about to be switched on and asked to move a safe distance from the rotating assembly, and will be informed of the potential risks (in particular, flying debris should the rotor or a 3D-printed part fail). Care will be taken to avoid scheduling testing during timeslots allocated to undergraduate practicals or training sessions in the workshop, and to keep the setup clear of other users' work areas. Where possible, testing will be coordinated with the lab engineer to avoid clashes with other booked activities.

General Housekeeping

The following housekeeping steps must be taken:

- All electrical connections will be insulated and continuity-tested to confirm that no loose or exposed connections exist before powering the setup.
- Any obstructions near and around the fan will be removed; the protective box/guard and safety glasses will be used to protect the operator and nearby users.
- After testing, the workspace will be cleared of any debris and disposed of appropriately.
- All bench DC supplies, soldering equipment and computers will be switched off at the wall and stored away.
- Tools and system components will be neatly packed into a labelled box, and placed into the respective locker in M214. Any damaged or broken components, tools or machinery will be reported to the lab engineer.
- Ensure no personal belongings are left behind and that the workspace is clean and the lab secured.

Fire Safety

The planned testing involves no direct open-flame fire risk. The fire-related risks are electrical (overheating wiring, short circuits, soldering). In the event of a fire emergency, refer to Appendix A for the emergency evacuation procedures and routes. The following fire-safety protocols apply:

- Earphones are prohibited in the laboratory and workshop so that the fire alarm remains clearly audible.
- All electrical connections will be checked and current-limited; overcurrent protection will be provided at 110 % of rated current.

- Soldering will be done only at a clean, open bench, away from other equipment, and the iron switched off after use.
- If the fire risk associated with the experiment is judged excessive by the responsible person, testing may be restricted to office hours.
- The LiPo battery will be disconnected at the end of every testing session, and its voltage will be verified with a multimeter before reconnecting to confirm it is within the safe operating range. The battery will be visually inspected before each use for signs of swelling, puncture, or damage to the casing, and will not be used if any such damage is present. Charging will only be carried out using a charger matched to the battery specification, on a non-flammable surface, and will never be left unattended while charging.

Activity Based Risk Assessment

Activity	Risk	Risk Type* (P/E)	Mitigating Steps	Classification of Risk Severity
Soldering	Solder-fume inhalation	P	Solder in a well-ventilated area and use fume extraction where available.	Possible
Soldering and electronics assembly	Burns from iron / molten solder	P/E	Solder only at a clean, open bench with no other equipment nearby, switch the iron off after use and use a stand.	Possible
Wiring & powering the DC supply	Electric shock	P/E	Inspect cable insulation before switching on, continuity-test all connections and never work on live circuits.	Possible
First start-up of the rotor	Instability / unexpected motion	P/E	Lab engineer or Head of Safety must approve before first start. Start at low rpm, guard in place, bystanders warned and clear.	Substantial
Increasing rotor speed to high rpm	Flying debris if a blade / printed part detaches	P/E	Protective box/guard around rotor with ≤ 10 mm openings. Safety glasses worn, tie up hair, no loose clothing. Increase speed gradually and keep hands clear.	High
Attaching LED arm and adding rotor electronics	Added rotating mass causing imbalance	P/E	Re-balance after each addition, re-verify stability at low speed before returning to operating speed.	Substantial
Viewing flashing LED display	Photosensitive - epilepsy trigger from strobe/flicker	P	Warn nearby users and avoid prolonged direct viewing. Limit exposure for anyone with photosensitive epilepsy.	Possible
Power outage / load shedding	Loss of data and uncontrolled stop	P/E	Check the load-shedding schedule, save data, switch equipment off at the wall when power is interrupted.	Possible

Data management	Loss of software and test data	P	Save work to a version-controlled (git) repository and keep independent backups.	Acceptable
Monitoring images and recording	Distraction near rotating equipment	P	Observe only through the guard opening, never reach into the guard while running and keep the test attended.	Acceptable
Charging the LiPo battery	Overheating or fire during charging	P/E	Charge on a non-flammable surface, away from other equipment, using a charger matched to the battery specification; do not leave charging unattended.	Possible
LiPo battery connection and operation	Fire, thermal runaway, or explosion from a damaged, or short-circuited cell	P/E	Inspect the battery for swelling or damage before use; verify voltage with a multimeter before connecting; disconnect at the end of every session; charge only with a matched charger and never leave charging unattended.	Substantial

*P – personal, E – equipment

Disciplinary Actions

Failure to comply with any of the aforementioned safety regulations or procedures will result in disciplinary action. Students will be issued an initial warning: after three warnings, the lab access is revoked for a month.

Appendix A: Emergency Evacuation Plans (M&M Training Workshop M214)

